S O L I T A R I O

Luxury Lab-Grown Diamond Jewellery | Diamond Education Series

Do Lab-Grown Diamonds Last Forever? What Happens After 10 Years.

A science-backed guide for millennial and Gen Z diamond buyers in India

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There is a question that comes up in almost every conversation about lab-grown diamonds, whispered in DMs, typed into search bars at midnight before a big purchase decision:

> *"But will it last? Will it still look the same in 10 years?"*

It is a completely valid question. You are investing in something meaningful — a ring that tells your love story, earrings that mark a milestone, a pendant that represents who you are. The last thing you want is for it to fade, cloud over, or lose its brilliance three years down the line.

The answer, backed by science and not just marketing: lab-grown diamonds are built to last a lifetime — and beyond. Here is everything you need to know.

What Are Lab-Grown Diamonds, Really?

Before we talk durability, let us settle one foundational truth: a lab-grown diamond is not a diamond simulant. It is not cubic zirconia. It is not moissanite. It is not glass with a sparkle filter.

A lab-grown diamond is a real diamond. Chemically, physically, and optically — it is identical to a mined diamond. Both are composed of pure carbon atoms arranged in a cubic crystal lattice structure, which is precisely what gives diamonds their legendary hardness and light-refracting brilliance.

The only difference is origin. Mined diamonds form over billions of years deep within the earth under extreme heat and pressure. Lab-grown diamonds replicate those exact conditions in a controlled environment — either through High Pressure High Temperature (HPHT) or Chemical Vapour Deposition (CVD) processes — completing the same journey in weeks rather than millennia.

Same carbon. Same crystal structure. Same diamond.

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The Science of Durability: Where Do Lab-Grown Diamonds Stand?

Hardness: The Mohs Scale

Hardness is measured on the Mohs scale, a 1–10 system that ranks minerals by scratch resistance. Diamond — whether mined or lab-grown — scores a perfect 10 out of 10. It is the hardest naturally occurring substance on Earth.

Material	Mohs Score	Scratch Resistance
Lab-Grown Diamond	10 / 10	Maximum — nearly unscratchable
Mined Diamond	10 / 10	Maximum — nearly unscratchable
Moissanite	9.25 / 10	Excellent but slightly softer
Sapphire / Ruby	9 / 10	Very good
Cubic Zirconia	8 – 8.5 / 10	Good, but chips over time
Glass	5.5 / 10	Poor — scratches easily

Structural Integrity

Beyond hardness, a diamond's toughness — its resistance to breaking or chipping — is also remarkable. Diamonds have a toughness rating of 7.5–8 MPa·m½, placing them among the most structurally robust gemstones available. Normal daily wear — including cooking, typing, and casual exercise — poses no meaningful threat to a well-set diamond.

Thermal and Chemical Stability

Lab-grown diamonds are chemically inert. They do not react to household chemicals, perfume, soap, chlorine, or sweat. They are not affected by heat from sunlight or cooking environments. The crystal structure that makes them so hard also makes them chemically stable across a wide range of everyday conditions.

> *A lab-grown diamond's Mohs hardness score of 10 means that only another diamond can scratch it. No daily activity, cleaning routine, or environmental condition you encounter in ordinary life will compromise its surface.*

What Actually Happens to a Lab-Grown Diamond After 10 Years?

Let us be concrete. After a decade of daily wear, here is what you can realistically expect from your lab-grown diamond:

The Diamond Itself

- The crystal structure remains completely unchanged — no degradation, no softening, no internal shifts.
- Surface brilliance is preserved, provided the diamond receives routine cleaning (warm water, mild soap, soft brush — once a month).

- No fading. No colour change. No cloudiness from within the stone itself.
- The diamond's refractive index — the property that creates its fire and sparkle — is permanent and will not diminish over time.

The Setting (What Needs Attention)

What does change over time is the metal setting — and this applies equally to mined and lab-grown diamonds. Prongs can wear down, especially in yellow gold. Rhodium plating on white gold may need renewal every 2–3 years. Clasps and hinges on bracelets may require maintenance.

This is standard jewellery maintenance — not a diamond problem. At Solitario, our pieces are crafted in 18K gold and platinum settings designed for longevity, and we recommend a professional jewellery check annually to ensure your setting is secure and as beautiful as day one.

The Myth of "Clouding"

Can a lab-grown diamond cloud over? No — not in the way many buyers fear. A real diamond cannot become internally cloudy over time. What buyers sometimes notice is surface build-up: a thin film of oil, soap residue, lotion, or skin cells that dulls the appearance temporarily. A simple cleaning restores full brilliance instantly. This is not the diamond changing — it is just the diamond being worn.

→ **Related:** How to Care for Your Solitario Diamond Jewellery

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Lab-Grown vs Mined Diamonds: A Durability Comparison

When it comes to physical durability, lab-grown and mined diamonds are scientifically equivalent. No gemological test — including those by GIA, IGI, or any independent laboratory — can distinguish a lab-grown diamond from a mined diamond based on durability or structural characteristics.

Where they differ: market perception and resale ecosystems — not physical performance.

Property	Lab-Grown Diamond	Mined Diamond
Chemical Composition	100% Carbon (C)	100% Carbon (C)
Crystal Structure	Cubic lattice	Cubic lattice
Mohs Hardness	10 / 10	10 / 10
Refractive Index	2.417 – 2.419	2.417 – 2.419
Fades Over Time?	No	No
Clouds Over Time?	No	No
Needs Cleaning?	Yes (routine)	Yes (routine)

Property	Lab-Grown Diamond	Mined Diamond
Resale Market	Growing rapidly	Established
Price Point	40–70% more accessible	Higher entry cost

What About Resale Value? A Transparent Conversation

This is where we choose honesty over marketing — because at Solitario, we believe in building trust.

Lab-grown diamonds, like most luxury goods, do not offer the same resale returns as mined diamonds in the current market. This is primarily because the cost of producing lab-grown diamonds has decreased significantly over the last decade, which means the secondary market has adjusted accordingly.

However, here is the more complete picture: mined diamonds also do not hold their value the way popular belief suggests. Most mined diamonds lose 20–40% of their retail value the moment they are purchased, due to retail markups. Neither category is a reliable liquid investment in the way gold or sovereign bonds are.

What lab-grown diamonds do offer instead is extraordinary value at the point of purchase — which means the diamond you buy for your engagement ring can be significantly larger, higher clarity, and better cut for the same budget. The emotional and aesthetic value is immediate and enduring.

At Solitario, we are transparent about this because we believe the right diamond purchase is one you make with your eyes open — and your heart fully invested.

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Common Myths About Lab-Grown Diamonds — Addressed Directly

Myth: Lab-Grown Diamonds Will Lose Their Sparkle

False. Sparkle is a function of cut quality and refractive index — both permanent properties of the diamond. A well-cut lab-grown diamond from Solitario will sparkle just as brilliantly on your 25th anniversary as it does on day one, provided it is kept clean.

Myth: Lab-Grown Diamonds Are Fake

False. The term 'fake diamond' applies to simulants like cubic zirconia or moissanite — materials that imitate a diamond's appearance but have a different chemical composition. Lab-grown diamonds are compositionally, crystallographically, and optically identical to mined diamonds. They are graded and certified by the same institutions using the same standards.

Myth: Lab-Grown Diamonds Are Not Certified

False. Lab-grown diamonds are certified by leading international gemological institutes including the IGI (International Gemological Institute) and GIA (Gemological Institute of America). Every Solitario diamond comes with full certification, documenting its 4Cs: carat, cut, colour, and clarity.

Myth: You Cannot Tell the Difference in Quality

The opposite is often true. Because lab-grown diamonds are produced at a lower cost per carat, buyers frequently achieve higher quality within the same budget — opting for better cut grades, higher clarity, or larger carat sizes than they could afford in mined diamonds.

> *At Solitario, the majority of our collection features VS clarity and above with Excellent or Very Good cut grades — because we believe the quality of your diamond should match the significance of the moment it marks.*

How to Keep Your Lab-Grown Diamond Looking Flawless for Decades

A lab-grown diamond does not need special care beyond standard diamond jewellery maintenance. Follow these practices and your Solitario piece will remain as brilliant as the day it was crafted:

- Clean monthly with warm water, a drop of mild dish soap, and a soft toothbrush. Rinse and pat dry.
- Remove jewellery before applying lotions, sunscreen, perfume, or hairspray — these leave residue that dulls the surface temporarily.
- Store pieces individually in soft pouches or lined boxes to prevent scratching between pieces.
- Visit a professional jeweller once a year for a setting inspection and professional clean.
- Avoid wearing your ring during heavy manual work, gym sessions, or activities involving significant impact.

At Solitario, we offer complimentary cleaning and annual check-up services for all pieces purchased from our collection, because we are invested in your diamond's story for the long term.

Frequently Asked Questions About Lab-Grown Diamond Durability

Q: Do lab-grown diamonds last forever?

> Yes. Lab-grown diamonds are physically and chemically identical to mined diamonds, with a Mohs hardness score of 10 — the maximum possible. They do not degrade, fade, or change

over time. With proper care, a lab-grown diamond will last indefinitely and can be passed down as a family heirloom.

Q: Do lab-grown diamonds fade or lose sparkle over time?

No. A lab-grown diamond cannot fade. Its sparkle is determined by its cut and refractive index — both permanent properties. Surface residue from daily wear can temporarily reduce brilliance, but a simple cleaning fully restores it. The diamond itself does not change.

Q: Are lab-grown diamonds as hard as real diamonds?

Yes. Lab-grown diamonds are real diamonds and score a perfect 10 on the Mohs hardness scale — identical to mined diamonds. This makes them the hardest gemstone available and highly resistant to scratching, chipping, and everyday wear.

Q: Do lab-grown diamonds cloud over time?

No. A lab-grown diamond will not cloud internally over time. What sometimes appears as cloudiness is surface buildup — oil, soap, or lotion residue — which is easily removed with a gentle clean. The internal crystal structure of a diamond cannot become cloudy.

Q: What is the resale value of lab-grown diamonds in India?

Lab-grown diamonds currently have a lower resale value than mined diamonds in the secondary market, as production costs have decreased. However, they offer exceptional value at purchase — often 40–70% more affordable than mined equivalents of the same quality, allowing buyers to invest in larger, higher-clarity stones. They are best considered as meaningful luxury purchases, not financial investments.

Q: How do I care for my lab-grown diamond jewellery long-term?

Clean your lab-grown diamond monthly with warm water, mild soap, and a soft brush. Store pieces individually, remove them before applying cosmetics or during heavy activity, and visit a jeweller annually for a professional inspection. This routine is identical to caring for any fine diamond jewellery.

Q: Are lab-grown diamonds certified in India?

Yes. Lab-grown diamonds sold by reputable jewellers in India — including Solitario — come with international certifications from bodies such as the IGI (International Gemological Institute) or GIA (Gemological Institute of America). These certificates detail the diamond's carat, cut, colour, and clarity grade.

The Verdict: Built to Last, Designed to Shine

Lab-grown diamonds do not just last — they endure. Through anniversaries, monsoons, decades of daily wear, and the inevitable adventures of a full life, the diamond you choose today will remain exactly as it is: brilliant, hard, and beautiful.

The question was never really about durability. It was about trust. Trust that your investment is real. Trust that your diamond is everything it claims to be. Trust that the piece you choose today will still take your breath away in 2035.

At Solitario, every lab-grown diamond in our collection is selected, certified, and crafted to honour that trust. We are not just selling jewellery — we are offering a piece of the future, worn with the confidence of forever.

Ready to choose your forever diamond?

Explore Solitario's certified lab-grown diamond jewellery collection — rings, earrings, pendants, and more.

About Solitario

Solitario is India's premier luxury lab-grown diamond jewellery brand. Every piece is IGI or GIA certified, ethically produced, and designed to last generations. Explore our collections online and experience the new standard of fine jewellery.

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